INCH-POUND
MIL-STD-2151A(SH)
23 July 2018

SUPERSEDING
MIL-STD-2151(SH)
25 July 1984

DEPARTMENT OF DEFENSE
TEST METHOD STANDARD

INCLINED LADDER TREAD TEST METHODS AND EQUIPMENT FOR WEAR, SLIP RESISTANCE, AND IMPACT



AMSC N/A FSC 2040

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Historically, ladder treads for shipboard use have not been tested, except for the type G tread in accordance with 804-0860041, which is fabricated of layered glass fiber reinforced plastic. The acquisition specification for the type G tread required the following tests for wear, slip resistance, and impact:

(a) The specified test for wear employed a rotating steel disk to which loose abrasive was applied. A 2- by 3-inch sample of the tread surface was forced against the disk, causing wear on the sample. The thickness change was then measured and related to the number of turns on the disk.

(b) To measure slip resistance, three worn samples were tested simultaneously by measuring the force required to pull a weight with three rubber feet across the samples for dry, wet, and oily conditions. The force was then converted to coefficient of friction.

(c) The impact test specified for the type G tread consisted of repeatedly dropping a 2-pound steel ball onto a 6- by 6-inch tread sample in a closely spaced pattern. The sample was then inspected for damage or loss of adhesion. Since most damage due to impact occurs on the nose of treads, the impact test did not simulate actual conditions.

3. It has been determined that tests for wear and slip resistance were unsatisfactory for the following reasons:

(a) The sample was honed to a flat, smooth surface, unlike actual shipboard wear caused by the scuffing action of shoe soles imbedded with abrasive dirt particles.

(b) The relative wear rates of different types of treads were not the same for lab tests as for shipboard tests.

(c) The test was conducted on a flat sample of tread without regard to the influence of the nose configuration on tread life and slip resistance.

(d) The method of measuring wear indicated average wear rate of the points measured and did not allow prediction of the tread life.

(e) Slip resistance was measured for the honed flat samples and did not reflect the slip resistance of a worn tread nose.

4. Tests have been incorporated into this standard that more closely simulate shipboard conditions and permit an evaluation of sample ladder treads with some degree of assurance that shipboard performance will be predicted.

5. Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH PAGE
1. SCOPE 1
1.1 Scope 1
2. APPLICABLE DOCUMENTS 1
2.1 General 1
2.2 Government documents. 1
2.2.1 Other Government documents, drawings, and publications 1
2.3 Order of precedence 1
3. DEFINITIONS 1
4. GENERAL REQUIREMENTS 1
4.1 Recycled, recovered, environmentally preferable, or biobased materials 1
4.2 General description 1
4.2.1 Wear test method 1
4.2.2 Slip resistance test method 2
4.2.3 Impact resistance test method 2
5. DETAILED REQUIREMENTS 2
5.1 Test equipment and accessories. 2
5.1.1 Description of wear test machine 2
5.1.2 Wear measuring accessories 2
5.1.3 Slip resistance measuring accessories 2
5.1.4 Impact equipment 2
5.2 Test methods. 2
5.2.1 Wear test procedure. 2
5.2.1.1 Tread mounting 2
5.2.1.2 Wear pad installation 2
5.2.1.3 Machine operation and care 3
5.2.1.3.1 Preparation 3
5.2.1.3.2 Counter 3
5.2.2 Wear measurement. 3
5.2.2.1 Measuring jig adjustment 3
5.2.2.2 Depth readings 3
5.2.2.3 Reading data 3
5.2.2.4 Duration and computing the average wear 3
5.2.3 Slip resistance measurement 3
5.2.3.1 Preparation and care of sole pads 3
5.2.3.2 Preparation of machine and accessory arrangement 4
5.2.3.3 Pad force 4
5.2.3.4 Slip resistance readings 4
5.2.3.4.1 Removal of loose material 4
5.2.3.4.2 Drying the tread surface 4
5.2.3.5 Duration and data recording 4
5.2.4 Impact tests 4
5.3 Test criteria. 5
5.3.1 Wear resistance 5
5.3.2 Slip resistance 5
5.3.3 Impact resistance 5
6. NOTES 5
6.1 Intended use 5
6.2 Acquisition requirements 5
6.3 Subject term (key word) listing 5
6.4 Changes from previous issue 5

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

LIST OF FIGURES

FIGURE PAGE
FIGURE 1. Data point locating jig. 6
FIGURE 2. Sole pad. 7
FIGURE 3. Tread impact test fixture. 8
FIGURE 4. Rigging for slip resistance test. 9
FIGURE 5. Tread positioning. 10
FIGURE 6. Clamp positioning. 10
FIGURE 7. Testing for pad pressure. 11
FIGURE 8. Example of completed record form. 12

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard describes test methods and equipment for testing compound-filled inclined ladder treads intended for use on Navy ships for wear, slip resistance, and impact resistance.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

804-0860041	-	Ladder, Inclined, Tread Details, Aluminum and Steel
804-5959201	-	Ladder Treads, Test Machine

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

This section is not applicable to this standard.

4. GENERAL REQUIREMENTS

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4.2 General description. Shipboard tests and examinations have shown that maximum tread wear and damage occur on the nose of the tread. Test methods for wear, slip resistance, and impact, as described herein, provide an approximate simulation of deterioration of inclined ladder treads encountered on board U.S. Navy ships.

4.2.1 Wear test method. The wear test is designed to provide a wear pattern of approximately the same contour and angle as for shipboard performance by drawing spring-loaded abrasive pads across the nose at an average angle of 30 degrees. The 3- by 6-inch pads are attached to a pair of articulated paddles carried by a rotating arm. The arm is rotated by a ⅓-horsepower (hp), 28.5 revolutions-per-minute (r/min) gear motor, causing 57 pad strokes per minute. The number of strokes are counted, permitting preset stroke count intervals as specified in the applicable acquisition document for measuring wear. Wear is measured at 17 positions in the contact area with a dial micrometer through holes in a jig plate. Wear rate is compared with the thickness of the slip-resistant material to indicate the life of tread wear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

4.2.2 Slip resistance test method. The slip resistance is measured at intervals as specified in the applicable acquisition document by applying a dry, a wet, and an oily rubber shoe sole pad to the wear pad and measuring the force required to pull the pad across the nose.

4.2.3 Impact resistance test method. Impact resistance is tested by dropping a 2-pound steel ball on various points of the tread flat and nose. The tread is then examined for cracking or loss of slip-resistant material.

5. DETAILED REQUIREMENTS

5.1 Test equipment and accessories.

5.1.1 Description of wear test machine. The wear test machine, as shown on 804-5959201, consists of a motor-driven twin leg stroke assembly. Each leg consists of an articulated, spring-loaded wear pad support. The 3- by 6-inch rubberized abrasive wear pad is mounted to the pad support with two-sided foam tape. A tread support fixture is provided to permit mounting of the tread at the required angle, height, and transverse position regardless of the tread design. A predetermining mechanical counter is mounted to record each revolution (two strokes per revolution) and to stop operation when the desired number of strokes is reached.

5.1.2 Wear measuring accessories. The wear measuring accessories consist of a data-point-locating jig, as shown on figure 1, and a depth dial micrometer, Starrett No. 644, or equal. The jig is designed to permit the perpendicular measurement of tread thickness changes at an angle of 30 degrees to the top of the tread at 17 points on the nose wear area. One end support for the jig is adjustable to eliminate rocking due to unevenness of the tread surface. Two additional holes in the jig are provided to permit reference readings to verify consistent jig positioning.

5.1.3 Slip resistance measuring accessories. Slip resistance is measured by pulling sole pads across the wear contact area and measuring the force required. Three sole pads (see figure 2) are used. Sole material is “Cats Paw”, or equal, 12-iron (¼-inch thick) rubber shoe sole material. Plain tap water is used for the wetting agent for the wet test condition, and SAE 10W-30 motor oil is applied for the oily condition. The force gage is a Chatillon Model DPP 50, or equal, with a scale of 0 to 50 pounds and a 0.5 pound graduation. The winch is a 1/135-horsepower by 12-revolutions-per-minute gear motor mounted as shown on 804-5959201 and fitted with a ⅝-inch diameter shaft extension to provide a tow speed of approximately 24 inches per minute. Tow rigging is one ⅛-inch diameter cotton rope loop and a small diameter flexible tow wire fitted with a hook for attachment to the force gage.

5.1.4 Impact equipment. Impact resistance is measured using the test fixture shown on figure 3. A 2-pound steel ball is dropped onto the tread from the height and at the locations specified. The fixture can be adjusted for variations in drop height, tread length, and impact angle. A plumb bob used for accurate position of the impact point and a hand-held vacuum release device (Neward Enterprises, Inc., Mitivac #6810, or equal) are used for dropping the ball. Full treads may be bolted directly to the fixture. When cap treads are tested, a type F extruded aluminum tread (see 804-0860041) is first installed, and the cap tread is then bolted or clamped to it. Foam padding is installed in the bottom of the fixture to catch the ball after impact.

5.2 Test methods.

5.2.1 Wear test procedure.

5.2.1.1 Tread mounting. The sample tread shall be mounted as shown on figure 4 using C-clamps. The mounting fixture shall be adjusted so that the tread is in the location and angle as shown on figure 5. Wood blocking or shims may be required when the tread is too narrow or has ribs which prevent solid clamping to the fixture. Transverse positioning is not critical. The clamps shall be located to also secure the positioning blocks for the wear measuring jig, as shown on figure 6.

5.2.1.2 Wear pad installation. The 3- by 6- by ⅜-inch rubberized abrasive wear pads shall be affixed to the face of the pad support with three strips of 1- by 1/16-inch, two-sided urethane foam tape along the length. Contact surfaces of the pad shall be thoroughly cleaned with soap and water and dried. Dirt and adhesive residue shall be removed from the pad support with solvent and a razor blade to assure proper adhesion. When the pads show signs of uneven wear or deep grooves, they shall be smoothed with a dressing stick (Brightboy #90, or equal) or replaced with a new pad. If the pad is worn to less than 3/16-inch thick in any part, it shall be replaced before the next tread test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

5.2.1.3 Machine operation and care. After connecting to a 115-volt outlet, the predetermining counter shall be set to half the number of strokes as specified (see 6.2). Pad force on the tread nose shall be checked to ensure proper spring tension by measuring the perpendicular force required to pull the pad away from the tread nose. The force gage and rope sling may be used for this purpose (see figure 7). If a force of 15 to 18 pounds is not obtained, adjustment of the spring length by modifying the mounting hooks, or replacement of the springs may be necessary. After any spring has been in service in excess of 50,000 strokes and the pad force falls below 15 pounds, the spring shall be replaced. Low force is an indication that the spring may fail from fatigue, causing personnel injury and machine damage.

5.2.1.3.1 Preparation. Before the wear test procedure, all set screws, mounting bolts, and clamps shall be tight, and arm return bumpers and flexible coupling inserts shall be in good condition. Lubrication of the gear motor is not required. When necessary, a drop of oil shall be placed on the pad arm pivot for smooth operation, and the pivot shaft nuts shall be tightened to reduce play.

5.2.1.3.2 Counter. When the thumb lever on the side of the counter is depressed all the way down and released, the machine will start and the counter will reset to the preset number. Caution shall be taken to stand clear while the machine is in operation. If necessary, the machine may be turned off during operation using the switch with the red cover plate. When using the counter lever, stopping and restarting will reset the counter and disrupt the stroke count.

5.2.2 Wear measurement.

5.2.2.1 Measuring jig adjustment. Before mounting the measuring jig on the tread, excessive tread surface variations shall be removed in the area of jig contact at each end of the jig with a coarse file. The jig shall be placed between the clamp blocks as described in 5.2.1.1 and as shown on figure 6. The end plate adjustment screw shall be loosened at one end and the end plate moved until both ends plates sit hard on the tread surface with no rocking. The screw shall be retightened.

5.2.2.2 Depth readings. Tread wear shall be determined by measuring the change in distance from the jig face to the tread surface by placing the depth dial micrometer through each of the holes in the jig and recording the reading as specified in 5.2.2.3. Data points “R” (see figure 1) shall be read first and shall be the same for each reading to verify consistent positioning of the jig. Contact surfaces between the jig and micrometer shall be clean to ensure accurate readings.

5.2.2.3 Reading data. Depth readings, rounded off to the nearest thousandth of an inch, shall be recorded. Date and initials of the person taking the data shall be recorded. Observations of unusual happenings or conditions shall also be recorded on the form.

5.2.2.4 Duration and computing the average wear. The duration of the wear-resistance test shall be 50,000 cycles. Depth readings shall be taken of the test sample through the 17 holes in the jig, as shown on figure 1, with a depth dial micrometer, before the initiation of the test and following the completion of 50,000 cycles. The depth readings shall be recorded in the left portion of the spaces, as shown on figure 8, corresponding to the hole location in the jig, as shown on figure 1. After the data have been recorded for the final stroke (50,000), the last readings shall be compared with corresponding initial readings and the differences recorded in the right portion of the appropriate spaces as shown on figure 8. The largest five differences in depth readings (as an example, in figure 8 the five largest differences in depth readings are 12, 13, 13, 14, and 16, which are in the cells with shaded upper right corners) recorded shall be averaged. The average wear (as an example, 0.014 in figure 8) shall be recorded in the lower right-hand cell, as shown on figure 8.

5.2.3 Slip resistance measurement. The slip-resistance measurement shall be made following the completion of the wear-resistance test.

5.2.3.1 Preparation and care of sole pads. Three soles pads, as shown on figure 2, shall be prepared by abrading the tread contact surface with 100 grit abrasive paper until the surface is thoroughly and uniformly abraded. Before use, foreign material shall be removed from the surface of the pads with a dry cotton cloth. The dry sole shall be used without further preparation. The wet sole shall be held under running tap water and rubbed to ensure thorough wetting. Excess water shall be removed and the wet sole applied directly to the tread. The oily sole contact surface shall be thoroughly coated with a thin coat of room-temperature SAE 10W-30 motor oil. The surface shall appear uniformly glossy. Before each test, the oily sole shall be cleaned of all foreign material and recoated with oil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

5.2.3.2 Preparation of machine and accessory arrangement. Rigging for the slip resistance test shall be as shown on figure 4 and prepared as follows:

a. Loosen the motor side of the drive shaft coupling and disengage the coupling.

b. Place the stop under the back side of the pad support arm, as shown on figure 7, to prevent reverse rotation of the assembly and properly position the sole pad on the tread.

c. Verify sole pad pressure as specified in 5.2.3.3.

d. Loop the rope sling over the pad support arm and hang the hook end of the force gage on it.

e. Hook the wire ladder from the winch to the screw on the back of the force gage.

5.2.3.3 Pad force. Pad force is measured in pounds. Using the rope sling and force gage, as shown on figure 7, the force required to pull the sole pad free of the tread surface shall be determined with the sole pad located at the starting position for the slip resistance test and rope sling centered on the tread nose. Pull shall be gently exerted on the gage until the sole pad lifts from the tread. This procedure shall be repeated three times. The average force shall be recorded (see figure 8).

5.2.3.4 Slip resistance readings. Slip resistance shall be measured in pounds. After rigging, as described in 5.2.3.2, the force gage indicator dial shall be set to "0" and the switch placed on the indicator next to the dial in the "lock" position. The tow winch shall be activated to pull the sole pad across the tread nose for approximately 1½ inches and then the winch shall be stopped. The dial will read the maximum force required to start movement of the sole pad. The rigging shall be disconnected, the sole pad returned to the start position, and the force gage reset. This process shall be repeated two more times. The average reading shall be recorded. If one reading is inconsistent by more than 2 pounds, the process shall be repeated until uniform readings are obtained. The slip-resistance factor is dimensionless and shall be determined by dividing the slip resistance (pounds) by the pad force (pounds).

5.2.3.4.1 Removal of loose material. Loose material shall be removed from the tread contact area with a stiff brush and the readings procedure repeated with the wet sole pad.

5.2.3.4.2 Drying the tread surface. The tread surface shall be dried with a cotton cloth and the readings procedure repeated for the oily sole pad. Upon completion of the slip-resistance tests, the tread contact surface shall be cleaned with a 95 percent solution of ethyl alcohol to remove all traces of oil before additional wear testing.

5.2.3.5 Duration and data recording. The duration of the slip-resistance test will be performed three times and the average measurement shall be recorded. Test data shall be recorded on the tread test record (see figure 8).

5.2.4 Impact tests. Impact tests shall follow the completion of the wear-resistance tests (4.2.1, 5.1.2, and 5.2.2) and the slip-resistance tests (4.2.2, 5.1.3, and 5.2.3). The impact test fixture (see figure 3) shall be used. Unless otherwise specified (see 6.2), the impact tests shall consist of dropping a 2-pound steel ball from a height of 72 inches onto the ladder tread. The tread shall be mounted horizontally. The ball shall be dropped onto the center of a tread 3 times.

a. Treads that are 6 inches wide and between 18 and 24 inches long with end plates may be mounted, as shown on figure 3, using ⅜-inch diameter bolts. One end of the mounting fixture is adjustable to accommodate various length treads. Additional mounting holes are provided to permit mounting of the tread at 30 and 90 degrees to horizontal for the angular nose impact tests. Mounting of 4- or 9-inch-wide treads will require modification of mounting hole spacing.

b. Cap treads shall be mounted directly on a type F aluminum tread (see 804-0860041) to simulate shipboard installations.

c. When the cap tread is in position, adjust the drop height, if necessary, by removing the pins at the rear of the ball locator support and moving the support up or down to the desired position. Drop height is measured from the impact point on the tread to the underside of the ball when held by the vacuum support cup.

d. Set the drop position by placing the bob support plug into the hole in the locator bar with the cord and plumb bob hanging freely. Adjust the cord length, if necessary, so that the bob clears the tread by approximately ¼ inch.

e. Adjust the impact position by moving the locator bar until the plumb bob stabilizes at the desired impact position on the tread.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

f. Remove the bob support plug from the hole, slide it to the side so that the cord enters the adjacent slot, then lift the bob up and stow it on the locator bar. This is done to avoid hitting the bob with the ball. Place the ball support cup in the locator hole and hold the ball by actuating the hand-held vacuum pump. Release the ball by opening the pump valve. The impact point shall be within ⅛ inch of the target, or the drop pattern shall be repeated in a different location. Inspect the impact point and record results such as surface crushing, hairline cracks, delamination, and dislodged pieces. Always describe the size and extent of the damage.

5.3 Test criteria.

5.3.1 Wear resistance. When tested in accordance with 4.2.1 and 5.1.2, and measured in accordance with 5.2.2, wear-resistance measurements shall not exceed 0.018 inch after 50,000 cycles.

5.3.2 Slip resistance. When tested in accordance with 4.2.2 and 5.1.3, and measured in accordance with 5.2.3 following the completion of the wear-resistance tests (4.2.1, 5.1.2, and 5.2.2), slip-resistance factors shall not exceed the following: 1.50 (dry), 1.25 (wet), and 1.0 (oily).

5.3.3 Impact resistance. When tested in accordance with 4.2.3, 5.1.4, and 5.2.4, there shall be no visible cracks, delamination, or dislodged pieces. Loss of filler shall not exceed 0.15 inch when measured with a depth micrometer.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document establishes standard test methods and test equipment for the purpose of ensuring that ladder treads manufactured for the U.S. Navy will meet performance expectations associated with heavy usage, harsh maritime environments, and battle damage.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Number of strokes (see 5.2.1.3).

c. Impact tests (see 5.2.4).

6.3 Subject term (key word) listing.

Depth

Grit

Pad

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

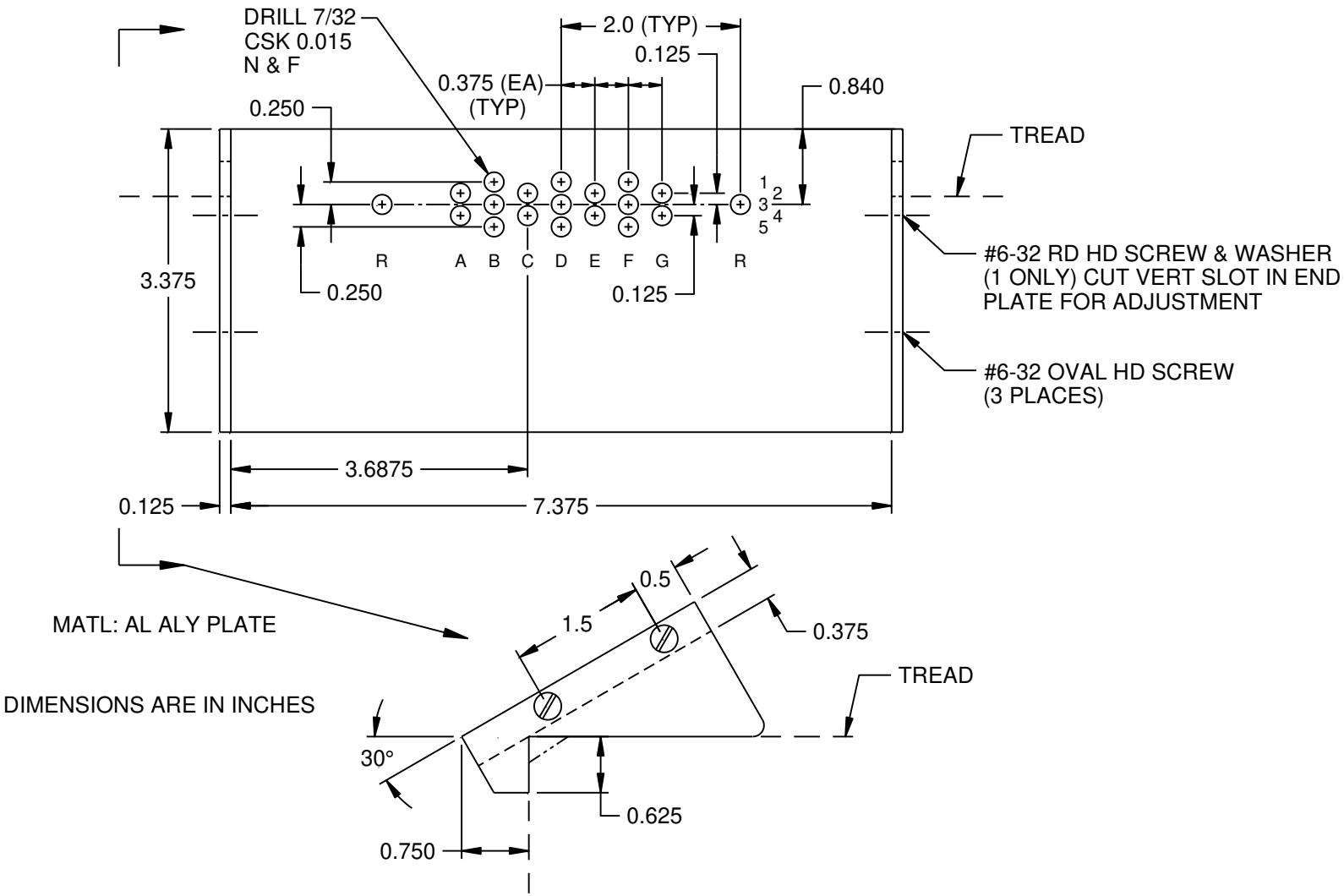
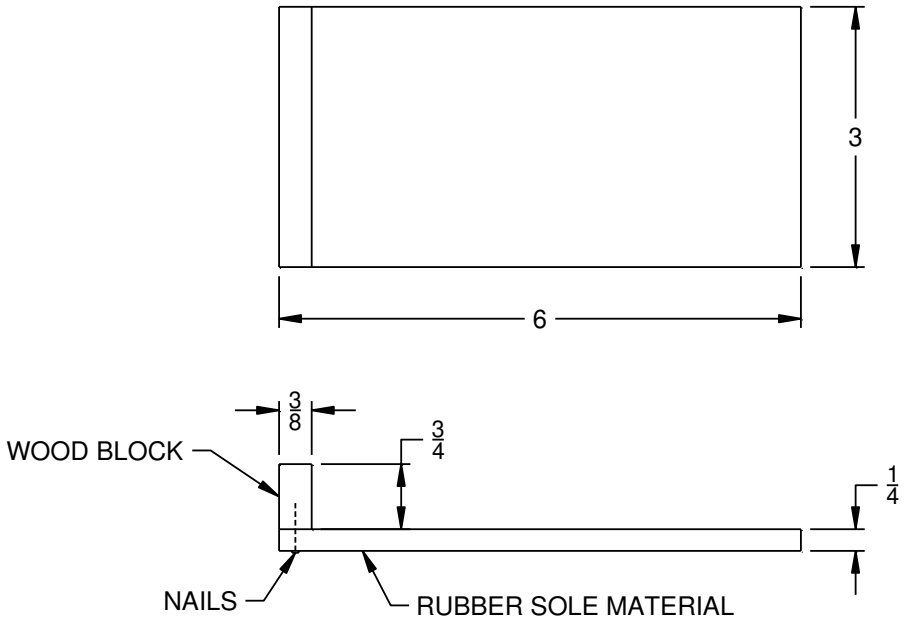


FIGURE 1. Data point locating jig.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.



DIMENSIONS ARE IN INCHES

FIGURE 2. Sole pad.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

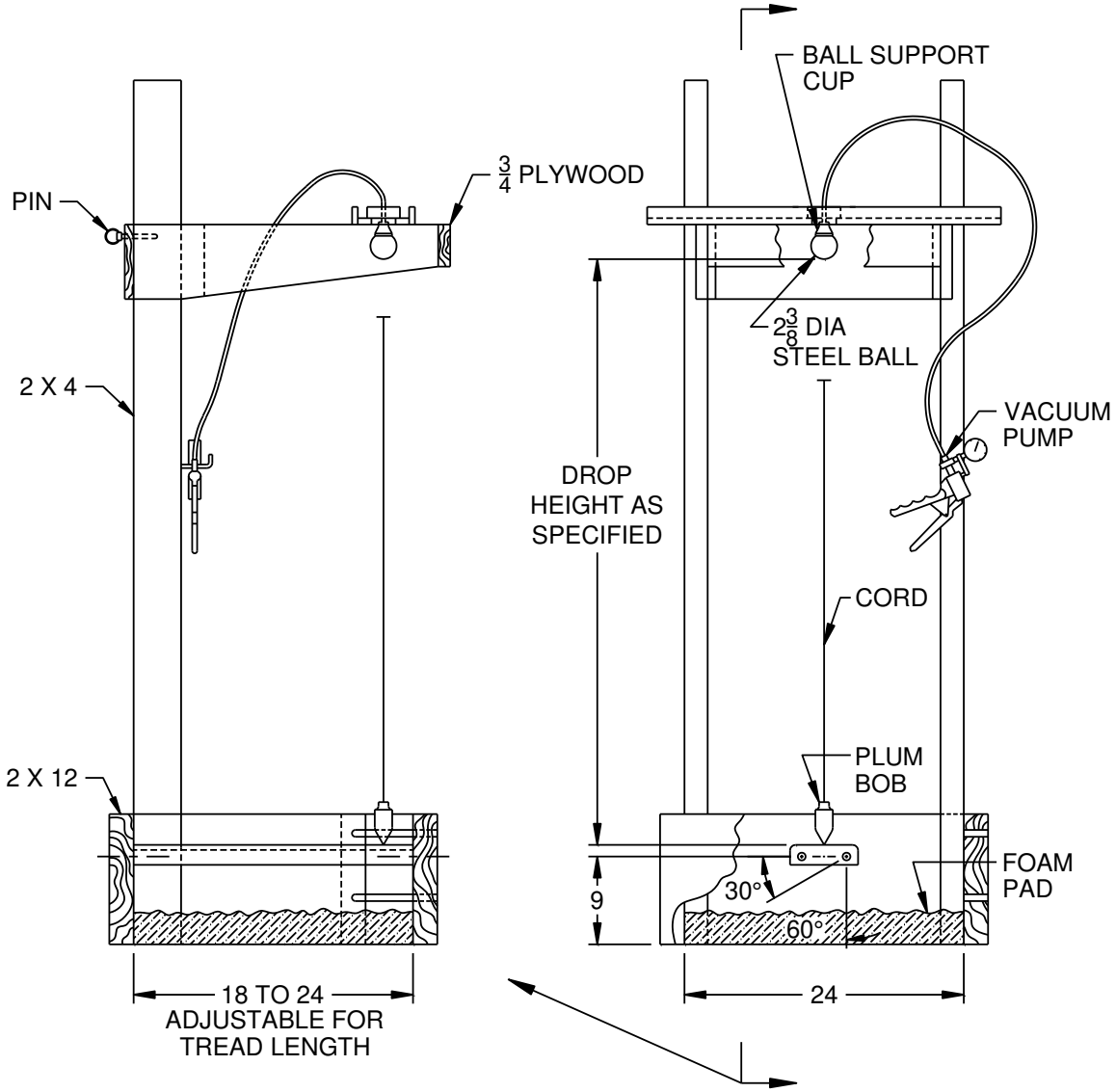


FIGURE 3. Tread impact test fixture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

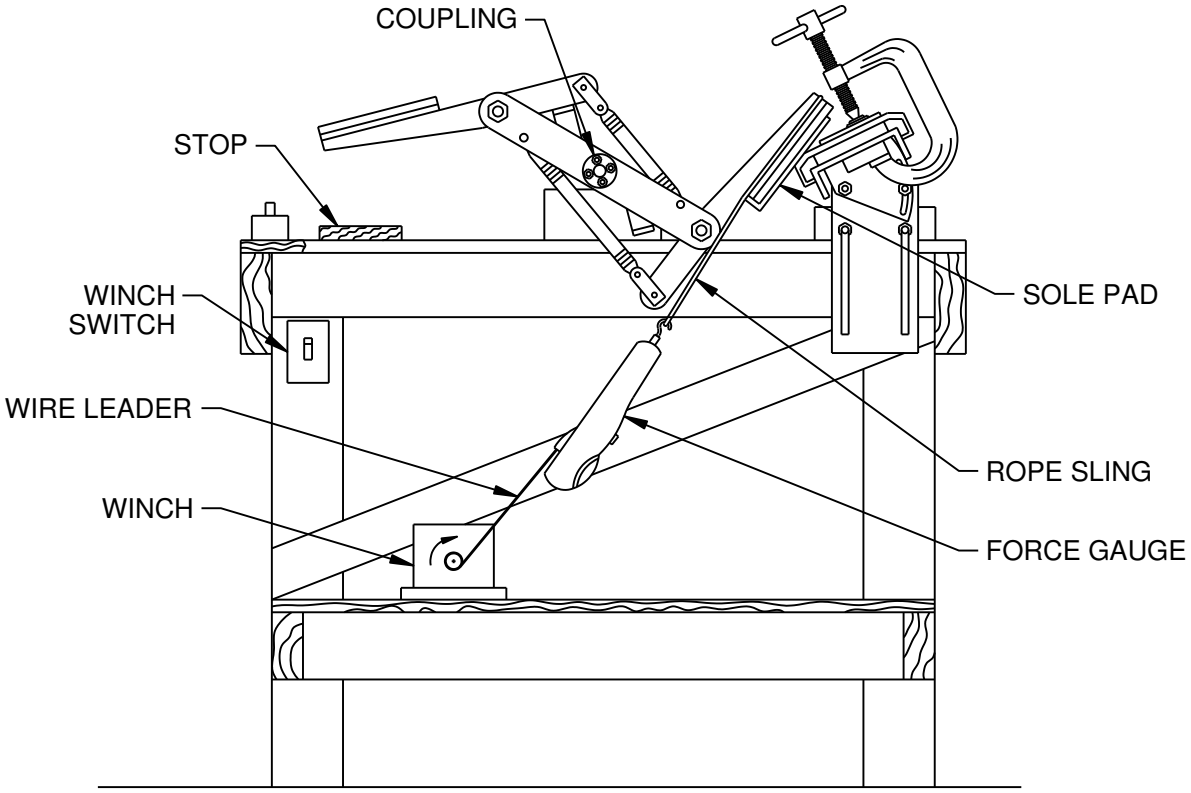


FIGURE 4. Rigging for slip resistance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

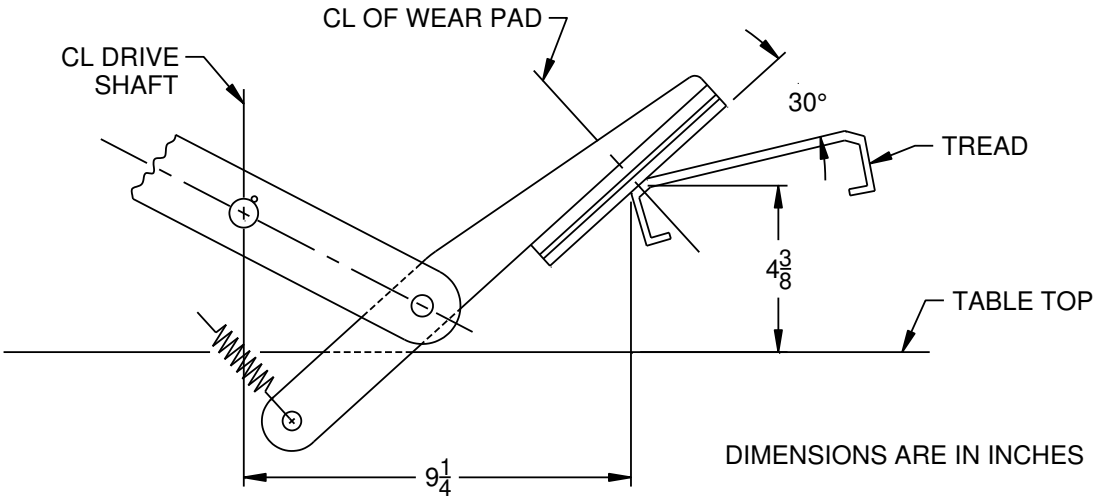


FIGURE 5. Tread positioning.

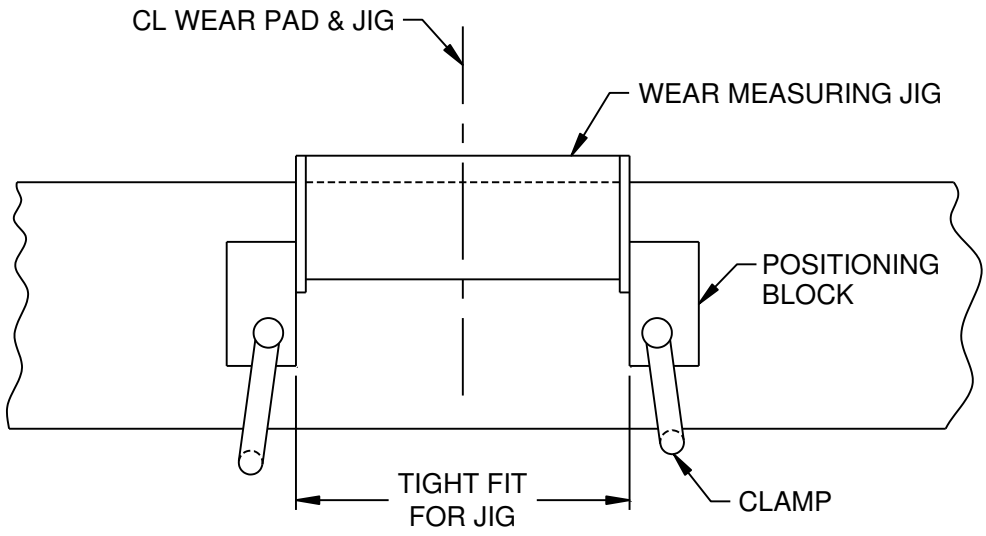


FIGURE 6. Clamp positioning.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

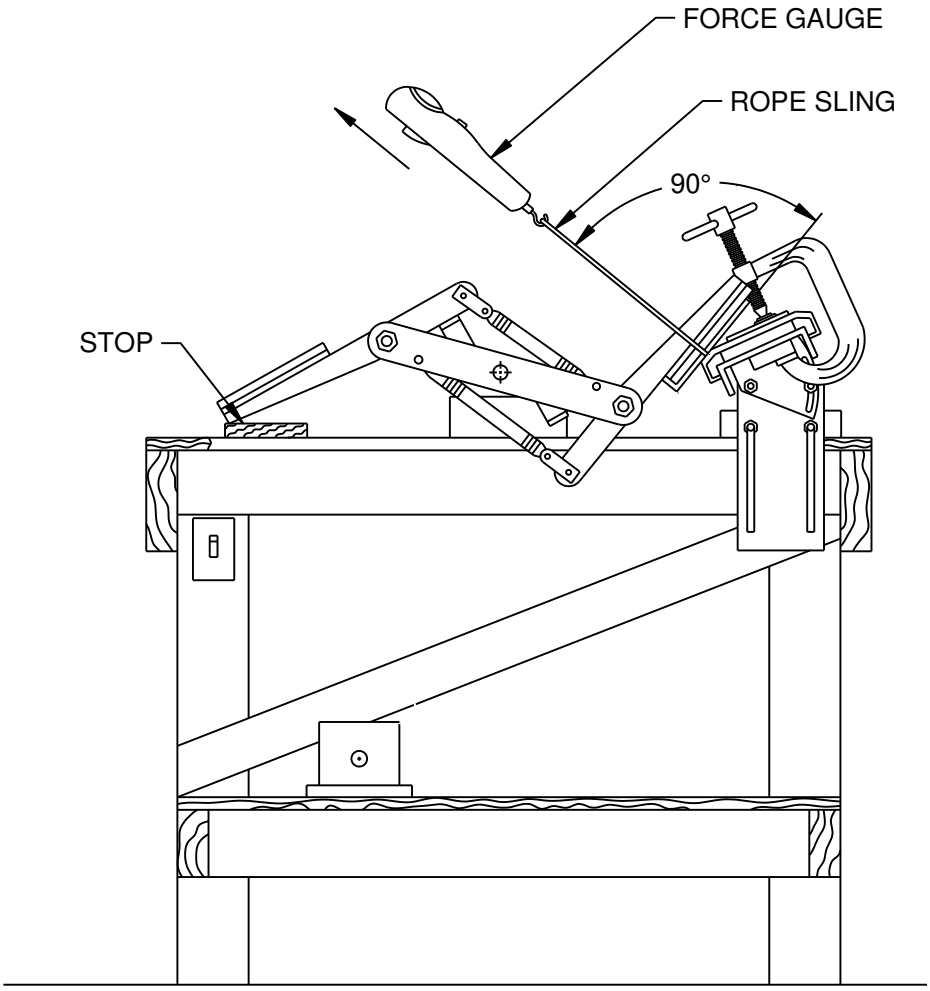


FIGURE 7. Testing for pad pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.

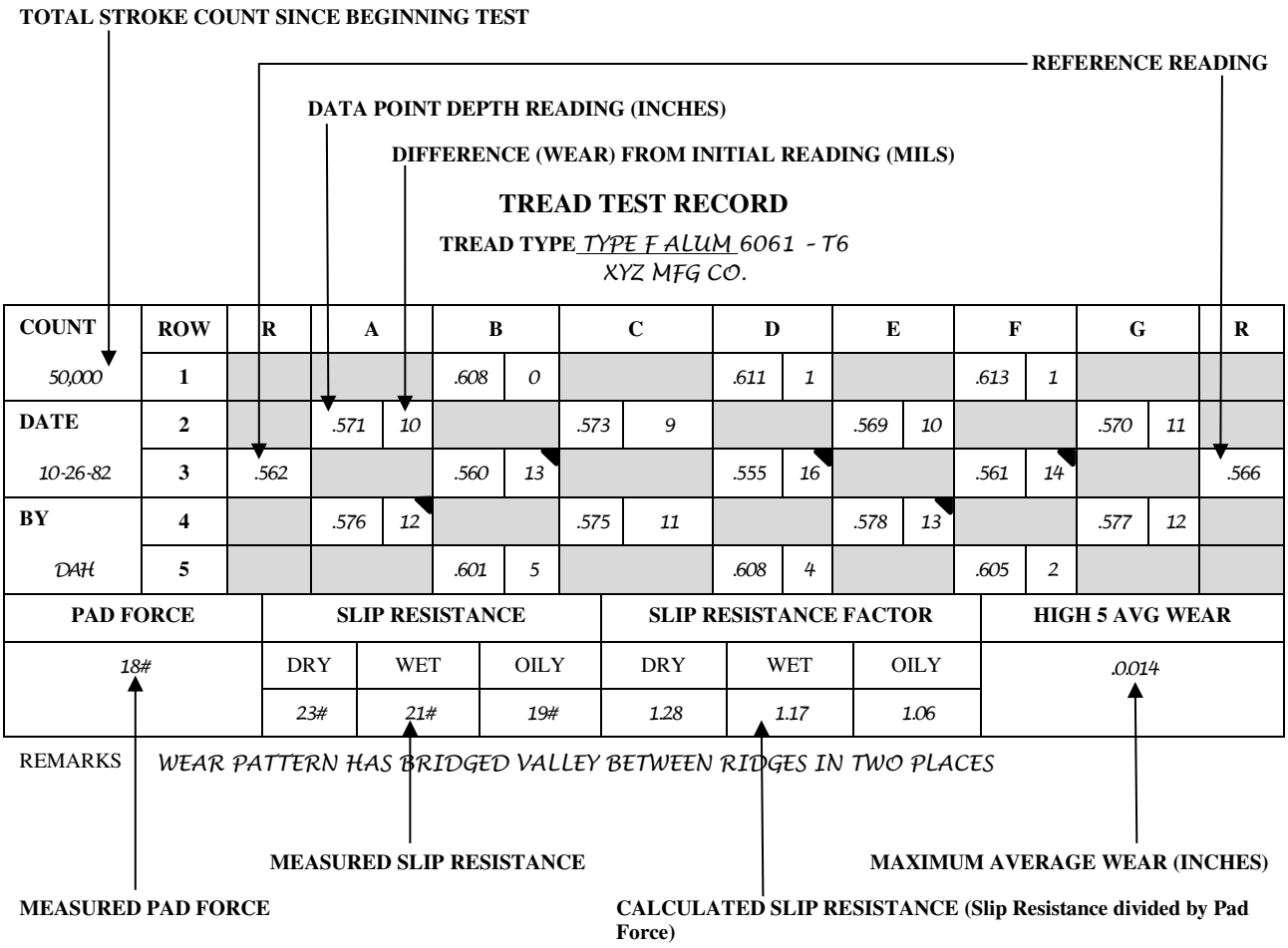


FIGURE 8. Example of completed record form.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 2040-2018-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:53Z
Check the source to verify that this is the current version before use.